

EMILIA LIM

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EDUCATION

SINGAPORE MANAGEMENT UNIVERSITY – Singapore

Aug 2021 – July 2025

Bachelor of Science (Information Systems): Major in Business Analytics

- **Relevant Coursework:** Data Management with SQL; Spreadsheet Modelling & Analytics; Visual Analytics for Business Intelligence; Data Analytics with R; Text Mining and Language Processing; Financial Analytics
- **Google Certifications:** Data Analytics, Advanced Data Analytics & Business Intelligence
 - Acquired expertise in data cleaning, analysis, visualisation, and predictive analytics, leveraging tools like SQL, R, and Tableau to derive actionable insights, create strategic dashboards, and support data-driven decision-making

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, HTML, CSS, JavaScript, PHP
- **Analytics & BI Tools:** Power BI, Tableau, Excel
- **Libraries & Frameworks:** Pandas, NumPy, scikit-learn, Matplotlib, Seaborn
- **Machine Learning & NLP:**
 - Models: Decision Tree, Random Forest, AdaBoost, Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), k-Nearest Neighbours, Naïve Bayes, Logistic Regression, Support Vector Machine, NLTK
 - Ensemble Methods: Stacking, Voting, Blending
- **Version Control & Deployment:** GitHub
- **Databases & Platforms:** MySQL
- **Other Tools:** Jupyter Notebook, Google Sheets, VS Code, Jira, Confluence

WORK EXPERIENCE

SingHealth – Singapore

June 2024 – Aug 2024

Data Science Intern

- **Objective:** To support clinical research and decision-making by analysing large-scale COPD patient data.
- Developed robust ETL pipelines in Python to process and analyse more than a million COPD-related clinical records.
- Applied data wrangling and transformation techniques to clean and analyse both structured and unstructured data.
- Developed data visualisations using Matplotlib and Seaborn to uncover patient-level patterns and support stakeholder reporting.
- Collaborated with respiratory experts, integrating clinical feedback to iterate on analysis and improve data quality and relevance.

SMU Academy – Singapore

May 2023 – Aug 2023

Project & Administrative Intern

- **Objective:** To improve course quality and trainer performance through data-driven monitoring and reporting.
- Collected, cleaned, and analysed course and trainer performance data using Excel and Python for quality monitoring.
- Built and maintained dashboards in Power BI to track KPIs, trainer effectiveness, and course engagement metrics.
- Delivered stakeholder reports with clear data storytelling and actionable recommendations.

Comfort Transportation Pte Ltd – Singapore

Mar 2022 – Jan 2023

Mobile App Tester

- **Objective:** To support quality mobile app releases through structured testing, issue reporting, and user-focused quality assurance.
- Responsible for cross-platform test planning and execution.
- Conducted testing for applications like CDG Zig and CDG Rentz in the UAT environment.
- Reported test results to the QA team and created bug tickets on Jira/ Confluence for product enhancement.
- Generated test scripts by formulating the steps needed to perform and test on specific user journeys.

PROJECTS

Early Diabetes Detection Using Machine Learning – [GitHub](#)

- **Objective:** To enhance early detection of diabetes through machine learning models that accurately identify at-risk individuals using health, demographic, and lifestyle data
- Built an ensemble-based predictive model to enhance early diabetes diagnosis using clinical, demographic, and lifestyle data.
- Trained and evaluated 10 machine learning algorithms: **Decision Tree, Random Forest, AdaBoost, XGBoost, LightGBM, k-Nearest Neighbours, Naïve Bayes, Logistic Regression, Support Vector Machine (SVM), Multi-Layer Perceptron (MLP)**
- Applied advanced preprocessing, handled class imbalance, prioritizing recall to minimize false negatives in medical diagnosis.
- Achieved strong performance with a final stacking ensemble (MLP, AdaBoost, LightGBM), reaching 85% recall, 85% precision and 93% AUC.

Credit Scorecard Analysis

- **Objective:** To assess credit risk and support lending decisions for Assisted Reproductive Technology (ART) loans.
- Built a credit scorecard for Assisted Reproductive Technology (ART) loans using logistic regression, Weight of Evidence (WOE), and Information Value (IV) to assess credit risk and support lending decisions.
- Performed data pre-processing, binning, and model evaluation to improve risk segmentation and support financial decision-making.

Climate Risk and Resilience Analysis

- **Objective:** To explore climate-related risks and resilience indicators across ASEAN nations.
- Developed a Tableau-based interactive dashboard to analyse and visualise climate risks, agricultural impacts, and energy trends in ASEAN countries, leveraging multi-source datasets including IMF, World Bank, and Oxfam reports.
- Utilised Python for data pre-processing (i.e. reshaping and cleaning high-dimensional climate datasets) and conducted exploratory data analysis (EDA) for visualisation design and storytelling.

Social Media Performance Analysis:

- **Objective:** Uncover audience sentiment and engagement trends to help a client optimise their social media strategy.
- Collaborated with a client to analyse the performance of their social media platforms.
- Conducted data scraping, cleaning, transformation, and text pre-processing techniques using Apify, Pandas, NumPy and NLTK.
- Utilised Matplotlib for visualisation and performed sentiment analysis using VADER.

Fraud Detection Analysis:

- **Objective:** Identify potential fraudulent emails by analysing linguistic patterns and enhancing detection accuracy through machine learning.
- Developed a scam detection model using the Enron fraud dataset, focusing on linguistic characteristics of emails using classification algorithms such as Naïve Bayes
- Applied text data cleaning techniques (tokenization, stop word removal, stemming/ lemmatization)
- and feature engineering to enhance model accuracy and data-driven decision-making.

ADDITIONAL

- Proficient in English and Mandarin (Written and Spoken)